

OBGEL Model Sheet: Cu–Ag – FCC A1

Open Binary Gibbs Energy Library

v0.2.0 candidate

Phase

The FCC A1 phase is represented as a substitutional solution model; below its miscibility gap, the same phase model supplies the Ag-rich and Cu-rich solid solutions.

Composition

The independent composition variable is $x = x_{\text{Cu}}$ with $x_{\text{Ag}} = 1 - x$.

Molar Gibbs free energy

$$\begin{aligned} G_m(x, T) = & xG_{\text{Cu}}^0(T) + (1 - x)G_{\text{Ag}}^0(T) \\ & + RT [x \ln x + (1 - x) \ln(1 - x)] \\ & + x(1 - x) \sum_k L_k(T)(2x - 1)^k. \end{aligned}$$

Redlich–Kister parameters

$$\begin{aligned} L_0(T) &= 36772.6 - 11.0285T, \\ L_1(T) &= 4612.43 - 0.28869T. \end{aligned}$$

Standard-state Gibbs energies

The pure-component functions are piecewise functions from the SGTE data for pure elements. The complete OBGEL model is limited by the common documented range of these functions.

Cu in FCC A1

$$\begin{aligned} G_{\text{Cu}}^0(T) = & -7770.458 + 130.48523T - 24.112392T \ln T - 0.00265684T^2 \\ & + 1.29223 \times 10^{-7}T^3 + 52478T^{-1} & 298.15 \leq T \leq 1357.77 \\ & - 13542.026 + 183.80383T - 31.38T \ln T + 3.642 \times 10^{29}T^{-9} & 1357.77 \leq T \leq 3200 \end{aligned}$$

Ag in FCC A1

$$\begin{aligned} G_{Ag}^0(T) = & -7209.512 + 118.20201T - 23.846331T \ln T - 0.001790585T^2 \\ & - 3.98587 \times 10^{-7}T^3 - 12011T^{-1} & 298.15 \leq T \leq 1234.93 \\ & - 15095.252 + 190.2664T - 33.472T \ln T + 1.412 \times 10^{29}T^{-9} & 1234.93 \leq T \leq 3000 \end{aligned}$$

Source and validation

The binary interaction parameters are reproduced from Kusoffsky (2002), which attributes them to a 1998 private communication from H. L. Lukas. NIST identifies the Hayes–Lukas–Effenberg–Petzow assessment as the full Cu–Ag thermodynamic assessment. Independent numerical evaluation of the complete Cu–Ag model gives a eutectic temperature of approximately 1053 K (779.9 C), consistent with the NIST value of 779.1 C.